QUASI-COMPACT IMAGE: CLOSED IFF SPECIALIZATION-STABLE

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Editorial note. Counted span is the author's own proof environment, source lines 3892-3906, following the statement at 3882-3890.

Allowed standard prerequisites (editorial).

- Closed sets are specialization-stable. Topology, Lemma topology-lemma-open-closed-specialization (cross-chapter, cited at ~3894). (1) implies (2).
- Image of a qc map stable under specializations. Topology, Lemma topology-lemma-lift-specializations-images (cross-chapter, cited at ~3930, used implicitly at ~3896). Intersection with affine open stays specialization-stable.
- Specialization-stable constructible image is closed. Algebra, Lemma algebra-lemma-image-stable-specialization-closed (cross-chapter, cited at ~3905). Concluding engine for the affine case.

Source reference. Original reference 00HY: focused-seeds/geometry/S9-schemes-image-quasi-compact-closed/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0062: focused-seeds/geometry/S9-schemes-image-quasi-compact-closed/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a quasi-compact morphism of schemes. The following are equivalent*

(1) *$f(X) \subset S$ is closed, and*
(2) *$f(X) \subset S$ is stable under specialization.*

Proof. We have (1) $\Rightarrow$ (2) by Topology, Lemma 0062. Assume (2). Let $U \subset S$ be an affine open. It suffices to prove that $f(X) \cap U$ is closed. Since $U \cap f(X)$ is stable under specializations in U, we have reduced to the case where S is affine. Because f is quasi-compact we deduce that $X = f^{-1}(S)$ is quasi-compact as S is affine. Thus we may write $X = \bigcup_{i=1}^{n} U_i$ with $U_i \subset X$ open affine. Say $S = \mathrm{Spec}(R)$ and $U_i = \mathrm{Spec}(A_i)$ for some R-algebra A_i. Then $f(X) = \mathrm{Im}(\mathrm{Spec}(A_1 \times \ldots \times A_n) \to \mathrm{Spec}(R))$. Thus the lemma follows from Algebra, Lemma 00HY. □

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